

Successive Differentiation.

If $y = f(x)$ is differentiable funⁿ.

$$\frac{dy}{dx} = f'(x) = y_1 \quad \text{--- } f^{\text{1st}} \text{ deriv.}$$

$$\frac{d^2y}{dx^2} = f''(x) = y_2$$

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$$\frac{d^ny}{dx^n} = f^{(n)}(x) = y_n$$

$$\textcircled{1} \quad y = e^{(ax+b)} \quad \left\{ \begin{array}{l} y_1 = e^{(ax+b)} (a) \\ y_2 = e^{(ax+b)} (a)^2 \\ \vdots \\ y_n = e^{(ax+b)} (a)^n \end{array} \right. \quad \textcircled{2} \quad y = a^{mx} \quad \left\{ \begin{array}{l} y_1 = a^{mx} \log a (m) \\ y_2 = a^{mx} (\log a)^2 (m)^2 \\ \vdots \\ y_n = a^{mx} (\log a)^n (m)^n \end{array} \right.$$

$$\textcircled{3} \quad y = (ax+b)^n \quad \left\{ \begin{array}{l} y_1 = n(ax+b)^{n-1} (a) \\ y_2 = n(n-1)(ax+b)^{n-2} a^2 \\ y_3 = n(n-1)(n-2)(ax+b)^{n-3} a^3 \\ \vdots \\ y_n = n(n-1)(n-2) \dots (n-(n-1)) (ax+b)^{n-n} a^n \end{array} \right. \quad \begin{array}{l} \textcircled{4} \\ 1 \quad 2 \quad 3 \quad 4 \\ 2 \quad 5 \cdot 4 \quad 2^2 \\ 3 \quad 5 \cdot 4 \cdot 3 \cdot 2^2 \\ 4 \quad 5 \cdot 4 \cdot 3 \cdot 2^3 \\ \vdots \\ 5 \quad 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \end{array}$$

$$\textcircled{5} \quad y = (ax+b)^{-m} \quad \left\{ \begin{array}{l} y_1 = (-m)(ax+b)^{-m-1} a \\ y_2 = (-m)(-m-1)(ax+b)^{-m-2} a^2 \\ y_3 = (-m)(-m-1)(-m-2)(ax+b)^{-m-3} a^3 \\ \vdots \\ y_n = (-m)(-m-1)(-m-2) \dots (-m-(n-1)) (ax+b)^{-m-n} a^n \end{array} \right. \quad \begin{array}{l} y_1 = 4(ax+b)^3 a \\ y_2 = 4 \cdot 3(ax+b)^2 a^2 \\ y_3 = 4 \cdot 3 \cdot 2(ax+b) a^3 \\ y_4 = 4 \cdot 3 \cdot 2 \cdot 1 a^4 \end{array}$$

$$\textcircled{6} \quad y = \log(ax+b) \quad \left\{ \begin{array}{l} y_1 = \frac{1}{(ax+b)} (a) \\ y_2 = (-1)(ax+b)^{-2} a^2 \\ y_3 = (-1)(-1)(ax+b)^{-3} a^3 \\ \vdots \\ y_n = (-1)(-1)(-1) \dots (-1)^{n-1} (ax+b)^{-n} a^n \end{array} \right. \quad \begin{array}{l} y_1 = (-1)(ax+b)^{-2} a^2 \\ y_2 = (-1)(-1)(ax+b)^{-3} a^3 \\ y_3 = (-1)(-1)(-1)(ax+b)^{-4} a^4 \\ \vdots \\ y_n = (-1)(-1)(-1) \dots (-1)^{n-1} (ax+b)^{-n} a^n \end{array}$$

$$\textcircled{7} \quad y = \sin(ax+b) \quad \left\{ \begin{array}{l} y_1 = \cos(ax+b) a \\ y_2 = -\sin(ax+b) a^2 \\ y_3 = -\cos(ax+b) a^3 \\ \vdots \\ y_n = (-1)^{n-1} \sin(ax+b) a^n \end{array} \right. \quad \begin{array}{l} y_1 = \cos(ax+b) a \\ y_2 = -\sin(ax+b) a^2 \\ y_3 = -\cos(ax+b) a^3 \\ \vdots \\ y_n = (-1)^{n-1} \sin(ax+b) a^n \end{array}$$

$$\textcircled{8} \quad y = \cos(ax+b) \quad \left\{ \begin{array}{l} y_1 = -\sin(ax+b) a \\ y_2 = \cos(ax+b) a^2 \\ y_3 = \sin(ax+b) a^3 \\ \vdots \\ y_n = (-1)^{n-1} \cos(ax+b) a^n \end{array} \right. \quad \begin{array}{l} y_1 = -\sin(ax+b) a \\ y_2 = \cos(ax+b) a^2 \\ y_3 = \sin(ax+b) a^3 \\ \vdots \\ y_n = (-1)^{n-1} \cos(ax+b) a^n \end{array}$$

$$\textcircled{9} \quad y = \tan(ax+b) \quad \left\{ \begin{array}{l} y_1 = \sec^2(ax+b) a \\ y_2 = 2 \tan(ax+b) \sec^2(ax+b) a^2 \\ y_3 = 2 \tan^2(ax+b) \sec^2(ax+b) a^3 \\ \vdots \\ y_n = 2 \tan^{n-1}(ax+b) \sec^2(ax+b) a^n \end{array} \right. \quad \begin{array}{l} y_1 = \sec^2(ax+b) a \\ y_2 = 2 \tan(ax+b) \sec^2(ax+b) a^2 \\ y_3 = 2 \tan^2(ax+b) \sec^2(ax+b) a^3 \\ \vdots \\ y_n = 2 \tan^{n-1}(ax+b) \sec^2(ax+b) a^n \end{array}$$

$$\frac{1}{(x-a)(x-b)} = \frac{1}{x-a} + \frac{1}{x-b}$$

$$\hookrightarrow \frac{1}{a+b} \quad \frac{1}{(a+b)^n}$$

$$\textcircled{1} \quad \frac{1}{(x-a)(x-b)(x-c)} = \frac{A}{(x-a)} + \frac{B}{(x-b)} + \frac{C}{(x-c)}$$

$$\textcircled{2} \quad \frac{1}{(x-a)^3(x-b)} = \frac{A}{(x-a)^3} + \frac{B}{(x-a)^2} + \frac{C}{(x-a)} + \frac{D}{x-b}$$

$$y = \frac{1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$$

$$= \frac{A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)}{(x-1)(x-2)(x-3)}$$

$$1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$$

$$x=1 \quad 1 = A(-1)(-2) \Rightarrow \boxed{A = \frac{1}{2}}$$

$$x=2 \quad 1 = B(1)(-1) \Rightarrow \boxed{B = -1}$$

$$x=3 \quad 1 = C(2) \Rightarrow \boxed{C = \frac{1}{2}}$$

$$y = \frac{1}{2} \frac{1}{(x-1)} - \frac{1}{(x-2)} + \frac{1}{2} \frac{1}{(x-3)}$$

$$\text{for } y = \frac{1}{ax+b} \quad y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

$$y_n = \frac{1}{2} \frac{(-1)^n n!}{(x-1)^{n+1}} - \frac{(-1)^n n!}{(x-2)^{n+1}} + \frac{1}{2} \frac{(-1)^n n!}{(x-3)^{n+1}}$$

$$y = \frac{x^2}{(x+2)(2x+3)} = \frac{x^2}{2x^2+7x+6}$$

$$\begin{array}{r} 2x^2+7x+6 \overline{) x^2} \\ \underline{x^2 + \frac{7}{2}x + 3} \\ -\frac{7}{2}x - 3 \end{array}$$

$$x^2 = (2x^2+7x+6) \frac{1}{2} + (-\frac{7}{2}x-3)$$

$$y = \frac{x^2}{(2x^2+7x+6)} = \frac{1}{2} + \frac{(-\frac{7}{2}x-3)}{2x^2+7x+6} \quad \text{--- (1)}$$

$$\frac{-\frac{7}{2}x-3}{(x+2)(2x+3)} = \frac{A}{x+2} + \frac{B}{2x+3} = \frac{A(2x+3) + B(x+2)}{(x+2)(2x+3)}$$

$$-\frac{7}{2}x-3 = A(2x+3) + B(x+2)$$

$$x = -2 \Rightarrow 4 = -A(1) \Rightarrow \boxed{A = -4}$$

$$x = -\frac{3}{2} \Rightarrow \frac{21}{4} - 3 = B(-\frac{3}{2}+2) \quad B = \frac{9}{2}$$

$$y = \frac{1}{2} + \frac{(-4)}{(x+2)} + \frac{9}{2} \frac{1}{(2x+3)}$$

$$\text{for } y = \frac{1}{ax+b} \quad y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

$$y_n = 0 - 4 \frac{(-1)^n n!}{(x+2)^{n+1}} + \frac{9}{2} \frac{(-1)^n n! 2^n}{(2x+3)^{n+1}}$$

$$y = \frac{1}{1+x+x^2+x^3}$$

$$-1, \pm i$$

$$= \frac{1}{(x+1)(x+i)(x-i)}$$

$$= \frac{A}{x+1} + \frac{B}{x+i} + \frac{C}{x-i}$$

$$= \frac{A(x+i)(x-i) + B(x+1)(x-i) + C(x+1)(x+i)}{(x+1)(x+i)(x-i)}$$

$$1 = A(x+i)(x-i) + B(x+1)(x-i) + C(x+1)(x+i)$$

$$x = -1 \Rightarrow 1 = A(1+i)(1-i) \Rightarrow 1 = A(2) \Rightarrow A = 1/2$$

$$x = -i \Rightarrow 1 = B(-i+1)(-2i) \Rightarrow 1 = -2(1+i)B \Rightarrow B = -\frac{1}{2(1+i)}$$

$$x = i \Rightarrow 1 = C(i+1)(2i) \Rightarrow 1 = 2(-1+i)C \Rightarrow C = \frac{1}{2(i-1)}$$

$$y = \frac{1}{2} \frac{1}{(x+1)} - \frac{1}{2(1+i)} \frac{1}{(x+i)} + \frac{1}{2(i-1)} \frac{1}{(x-i)}$$

$$\text{for } y = \frac{1}{ax+b} \quad y_n = \frac{(-1)^n n! (a)^n}{(ax+b)^{n+1}}$$

$$y_n = \frac{1}{2} \frac{(-1)^n n!}{(x+1)^{n+1}} - \frac{1}{2(1+i)} \frac{(-1)^n n!}{(x+i)^{n+1}} + \frac{1}{2(i-1)} \frac{(-1)^n n!}{(x-i)^{n+1}}$$

$$y = \frac{8x}{x^3 - 2x^2 - 4x + 8} \quad -2, 2, 2$$

$$= \frac{8x}{(x-2)^2(x+2)} = \frac{A}{(x-2)^2} + \frac{B}{x-2} + \frac{C}{x+2}$$

$$= \frac{A(x+2) + B(x-2)(x+2) + C(x-2)^2}{(x-2)^2(x+2)}$$

$$8x = A(x+2) + B(x-2)(x+2) + C(x-2)^2$$

$$x=2 \Rightarrow 16 = A \cdot 4 \Rightarrow \boxed{A=4}$$

$$x=-2 \Rightarrow -16 = C \cdot 16 \Rightarrow \boxed{C=-1}$$

$$x=0 \Rightarrow 0 = 2A - 4B + 4C$$

$$0 = 8 - 4B - 4 \Rightarrow 4B = 4$$

$$\boxed{B=1}$$

$$y = \frac{4}{(x-2)^2} + \frac{1}{x-2} - \frac{1}{x+2}$$

$$\text{for } y = \frac{1}{(ax+b)^m} = \frac{(-1)^n (m+n-1)! a^n}{(m-1)! (ax+b)^{m+n}} \quad \checkmark$$

$$y = \frac{1}{ax+b} = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

$$m=2$$

$$y_n = \frac{4(-1)^n (n+1)!}{(x-2)^{n+2}} + \frac{(-1)^n n!}{(x-2)^{n+1}} - \frac{(-1)^n n!}{(x+2)^{n+1}}$$

$$y = \frac{x}{(x+1)^4} = \frac{(x+1) - 1}{(x+1)^4}$$

$$= \frac{1}{(x+1)^3} - \frac{1}{(x+1)^4}$$

$p-1$. $\frac{x^3}{x^2-1}$ for $x=0$ is 0 if n is even
is $-n!$ if n is odd & > 1 .

$$y = \frac{x^3}{x^2-1} = \frac{x^3 - x + x}{x^2-1}$$

$$= x \frac{(x^2-1) + 1}{x^2-1}$$

$$= x + \frac{x}{x^2-1}$$

$$= x + \frac{x}{(x-1)(x+1)}$$

$$\frac{x}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1} = \frac{A(x+1) + B(x-1)}{(x-1)(x+1)}$$

$$\begin{aligned} x=1 & \quad 1 = 2A \Rightarrow A = 1/2 \\ x=-1 & \quad -1 = -2B \Rightarrow B = +1/2 \end{aligned}$$

$$y = x + \frac{1}{2} \left[\frac{1}{x-1} + \frac{1}{x+1} \right]$$

$$f_n = 0 + \frac{1}{2} \left[\frac{(-1)^n n!}{(x-1)^{n+1}} + \frac{(-1)^n n!}{(x+1)^{n+1}} \right]$$

$$y = \frac{1}{ax+b}$$

$$y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

$$\frac{(-1)^n}{(-1)^{n+1}(-1)}$$

put $x=0$

$$f_n(0) = \frac{(-1)^n n!}{2} \left[\frac{1}{(-1)^{n+1}} + \frac{1}{1} \right]$$

$$= \frac{n!}{2} \left[\frac{(-1)^n}{(-1)^{n+1}} + (-1)^n \right] = \frac{n!}{2} [-1 + (-1)^n]$$

if n is even

$$f_n(0) = \frac{n!}{2} [-1 + 1] = 0$$

if n is odd & > 1

$$f_n(0) = \frac{n!}{2} [-1 - 1] = -n!$$

$$y = x \log\left(\frac{x-1}{x+1}\right) \quad \text{p.t.} \quad y_n = (-1)^{n-2} (n-2)! \left[\frac{x-n}{(x-1)^n} - \frac{x+n}{(x+1)^n} \right]$$

$$y = x [\log(x-1) - \log(x+1)]$$

$$= x \log(x-1) - x \log(x+1)$$

$$y_1 = \frac{x}{x-1} + \log(x-1) - \left(\frac{x}{x+1} + \log(x+1) \right)$$

$$= \frac{x-1+1}{x-1} + \log(x-1) - \left(\frac{x+1-1}{x+1} + \log(x+1) \right)$$

$$= \cancel{x} + \frac{1}{x-1} + \log(x-1) - \left(\cancel{x} - \frac{1}{x+1} \right) - \log(x+1)$$

$$y_1 = \frac{1}{x-1} + \log(x-1) + \frac{1}{x+1} - \log(x+1)$$

$$\begin{matrix} y_1 & y_1 \\ y_1 & -y_2 \\ y_n & y_{n-1} \end{matrix}$$

$$\left. \begin{array}{l} y = \frac{1}{ax+b} \\ y = \log(ax+b) \end{array} \right\} \begin{array}{l} y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}} \\ y_n = \frac{(-1)^{n-1} (n-1)! a^n}{(ax+b)^n} \end{array} \quad \begin{array}{l} y_1 = y \\ y_2 = y_1 \\ y_n = y_{n-1} \end{array}$$

$$y_n = \frac{(-1)^{n-1} (n-1)!}{(x-1)^n} + \frac{(-1)^{n-2} (n-2)!}{(x-1)^{n-1}} + \frac{(-1)^{n-1} (n-1)!}{(x+1)^n} - \frac{(-1)^{n-2} (n-2)!}{(x+1)^{n-1}}$$

$$= \frac{(-1)^{n-2} (n-2)!}{(x-1)^n} \left[\frac{(-1)(n-1)}{(x-1)^n} + \frac{1}{(x-1)^{n-1}} + \frac{(-1)(n-1)}{(x+1)^n} - \frac{1}{(x+1)^{n-1}} \right]$$

$$= (-1)^{n-2} (n-2)! \left[\frac{-n+1}{(x-1)^n} + \frac{(x-1)}{(x-1)^n} + \frac{-n+1}{(x+1)^n} - \frac{(x+1)}{(x+1)^n} \right]$$

$$= (-1)^{n-2} (n-2)! \left[\frac{x-n}{(x+1)^n} - \frac{(x+n)}{(x+1)^n} \right] \quad \checkmark$$

$$\sin 2x = 2 \sin x \cos x$$

$$1 + \cos 2x = 2 \cos^2 x$$

$$1 - \cos 2x = 2 \sin^2 x$$

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$e^{\pm i\theta} = \cos \theta \pm i \sin \theta$$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\cosh \theta = \left(\frac{e^{\theta} + e^{-\theta}}{2} \right)$$

$$\begin{aligned}
 \textcircled{1} \quad y &= \cos x \cos 2x \cos 3x \\
 &= \cos x \left(\frac{\cos 5x + \cos(-x)}{2} \right) \\
 &= \frac{1}{2} \left[\cos x \cos 5x + \cos x \cos x \right] \\
 &= \frac{1}{2} \left[\frac{\cos 6x + \cos(-4x)}{2} + \frac{\cos 2x + \cos(0)}{2} \right] \\
 &= \frac{1}{4} \left[\cos 6x + \cos 4x + \cos 2x + 1 \right] \\
 y &= \cos(ax+b) \quad y_n = a^n \cos\left(\frac{n\pi}{2} + ax + b\right)
 \end{aligned}$$

$$y_n = \frac{1}{4} \left[6^n \cos\left(\frac{n\pi}{2} + 6x\right) + 4^n \cos\left(\frac{n\pi}{2} + 4x\right) + 2^n \cos\left(\frac{n\pi}{2} + 2x\right) + 0 \right]$$

$$\begin{aligned}
 y &= \sin^2 x \cos^3 x \\
 &= \frac{(\sin x \cos x)^2}{\left(\frac{\sin 2x}{2}\right)^2} \cos x
 \end{aligned}$$

$$1 - \cos 2x = 2\sin^2 x$$

$$= \frac{1}{4} (\sin^2 2x) \cos x$$

$$= \frac{1}{4} \left(\frac{1 - \cos 4x}{2} \right) \cos x$$

$$= \frac{1}{8} (1 \cos x - \cos 4x \cos x)$$

$$= \frac{1}{8} \left[\cos x - \frac{(\cos 5x + \cos 3x)}{2} \right]$$

$$= \frac{1}{16} \left[2 \cos x - \cos 5x - \cos 3x \right]$$

$$y_n = \frac{1}{16} \left[2 \cos\left(\frac{n\pi}{2} + x\right) - 5^n \cos\left(\frac{n\pi}{2} + 5x\right) - 3^n \cos\left(\frac{n\pi}{2} + 3x\right) \right]$$

$$y = \sin^4 x \quad \longrightarrow \quad (\sin^2 x)^2$$

$$= (\sin x)^4$$

$$= \left(\frac{e^{ix} - e^{-ix}}{2i} \right)^4$$

$$= \frac{1}{(2i)^4} (e^{ix} - e^{-ix})^4$$

$$= \frac{1}{16} [(e^{ix})^4 - 4(e^{ix})^3(e^{-ix}) + 6(e^{ix})^2(e^{-ix})^2 - 4(e^{ix})(e^{-ix})^3 + (e^{-ix})^4]$$

$$= \frac{1}{16} [e^{i4x} - 4e^{i2x} + 6 - 4e^{-i2x} + e^{-i4x}]$$

$$= \frac{1}{16} [e^{i4x} + e^{-i4x} - 4(e^{i2x} + e^{-i2x}) + 6]$$

$$= \frac{1}{16} [2\cos 4x - 4(2\cos 2x) + 6]$$

$$y = \frac{2}{16} [\cos 4x - 4\cos 2x + 3]$$

$$y_n = \frac{1}{8} [4^n \cos\left(\frac{n\pi}{2} + 4x\right) - 4 \cdot 2^n \cos\left(\frac{n\pi}{2} + 2x\right) + 0]$$

$$\left. \begin{aligned} y &= \cos(ax+bx) \\ y_n &= a^n \cos\left(\frac{n\pi}{2}, ax+bx\right) \end{aligned} \right\}$$

$$\sin^5 x \quad / \quad \cos^4 2x$$

$$\frac{e^{i\theta} + e^{-i\theta}}{2} = \cos \theta$$

$$\begin{array}{ccccccc} & & & & 1 & & \\ & & & & 1 & 2 & 1 \\ & & 1 & & 3 & & 3 & & 1 \\ & 1 & & 4 & & 6 & & 4 & & 1 \end{array} \quad \checkmark$$

9/6 $y = \sin rx + \cos rx$ then P.T.

$$y_n = r^n [1 + (-1)^n \sin 2rx]^{1/2} \quad \checkmark$$

$$y_n = r^n \sin\left(\frac{n\pi}{2} + rx\right) + r^n \cos\left(\frac{n\pi}{2} + rx\right)$$

$$= r^n \left\{ \left[\sin\left(\frac{n\pi}{2} + rx\right) + \cos\left(\frac{n\pi}{2} + rx\right) \right]^2 \right\}^{1/2}$$

$$= r^n \left[\sin^2\left(\frac{n\pi}{2} + rx\right) + \cos^2\left(\frac{n\pi}{2} + rx\right) + 2 \sin\left(\frac{n\pi}{2} + rx\right) \cos\left(\frac{n\pi}{2} + rx\right) \right]^{1/2}$$

$$= r^n \left[1 + \sin 2\left(\frac{n\pi}{2} + rx\right) \right]^{1/2}$$

$$= r^n [1 + \sin(n\pi + 2rx)]^{1/2}$$

$$= r^n [1 + (-1)^n \sin 2rx]^{1/2}$$

$$\begin{aligned} & \sin(n\pi + 2rx) \\ &= \sin n\pi \cos 2rx \\ & \quad + \cos n\pi \sin 2rx \\ &= 0 + (-1)^n \sin 2rx \end{aligned}$$

$$y = \frac{1}{x^2 + a^2} \quad \text{P.T.} \quad y_n = \frac{(-1)^n n!}{a^{n+2}} \sin^{n+1} \theta \sin(n+1)\theta$$

where $\theta = \tan^{-1}(a/x)$

$$y = \frac{1}{(x-ai)(x+ai)} = \frac{A}{x-ai} + \frac{B}{x+ai}$$

$$x^2 + a^2 = 0$$

$$x = \pm ai$$

$$1 = A(x+ai) + B(x-ai)$$

$$x = ai \quad A = \frac{1}{2ai}$$

$$x = -ai \quad B = -\frac{1}{2ai}$$

$$y = \frac{1}{2ai} \left[\frac{1}{x-ai} - \frac{1}{x+ai} \right]$$

$$\left| \begin{aligned} &(\cos \theta + i \sin \theta)^n \\ &= \cos n\theta + i \sin n\theta \end{aligned} \right.$$

$$y = \frac{1}{ax+b} \quad y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

$$y_n = \frac{1}{2ai} \left[\frac{(-1)^n n!}{(x-ai)^{n+1}} - \frac{(-1)^n n!}{(x+ai)^{n+1}} \right]$$

$$= \frac{(-1)^n n!}{2ai} \left[\frac{1}{(x-ai)^{n+1}} - \frac{1}{(x+ai)^{n+1}} \right]$$

put $x = r \cos \theta, a = r \sin \theta$

$$\tan \theta = \frac{a}{x} \Rightarrow \theta = \tan^{-1}(a/x)$$

$$= \frac{(-1)^n n!}{2ai} \left[\frac{1}{(r \cos \theta - i r \sin \theta)^{n+1}} - \frac{1}{(r \cos \theta + i r \sin \theta)^{n+1}} \right]$$

$$= \frac{(-1)^n n!}{2ai r^{n+1}} \left[(\cos \theta - i \sin \theta)^{-(n+1)} - (\cos \theta + i \sin \theta)^{-(n+1)} \right]$$

$$= \frac{(-1)^n n!}{2ai r^{n+1}} \left[\cancel{\cos(n+1)\theta} + i \sin(n+1)\theta - (\cancel{\cos(n+1)\theta} - i \sin(n+1)\theta) \right]$$

$$\sin(+n+1)\theta = \sin(n+1)\theta$$

$$\cos(-n+1)\theta = \cos(n+1)\theta$$

$$= \frac{(-1)^n n!}{2ai r^{n+1}} 2i \sin(n+1)\theta$$

$$\text{as } a = r \sin \theta \Rightarrow r = \frac{a}{\sin \theta}$$

$$= \frac{(-1)^n n!}{a \frac{a^{n+1}}{(\sin \theta)^{n+1}}} \sin(n+1)\theta = \frac{(-1)^n n!}{a^{n+2}} \sin^{n+1} \theta \sin(n+1)\theta$$

$$y = \tan^{-1} x \quad \text{p.t. } y_n = (-1)^n (n-1)! \sin^n \theta \sin n\theta$$

$$y_1 = \frac{1}{1+x^2} \quad \text{where } \theta = \tan^{-1}\left(\frac{1}{x}\right)$$

$$= \frac{1}{(x-i)(x+i)}$$

$$= \frac{A}{x-i} + \frac{B}{x+i}$$

$$1 = A(x+i) + B(x-i)$$

$$x=i \quad A = \frac{1}{2i}$$

$$x=-i \quad B = -\frac{1}{2i}$$

$$y_1 = \frac{1}{2i} \left[\frac{1}{x-i} - \frac{1}{x+i} \right]$$

$$y_n = \frac{1}{2i} \left[\frac{(-1)^{n-1} (n-1)!}{(x-i)^n} - \frac{(-1)^n (n-1)!}{(x+i)^n} \right]$$

$$x = r \cos \theta \quad 1 = r \sin \theta$$

$$\theta = \tan^{-1}\left(\frac{1}{x}\right)$$

$$= \frac{(-1)^n (n-1)!}{2i} \left[\frac{1}{(r \cos \theta - i r \sin \theta)^n} - \frac{1}{(r \cos \theta + i r \sin \theta)^n} \right]$$

$$= \frac{(-1)^n (n-1)!}{2i r^n} \left[(\cos \theta - i \sin \theta)^{-n} - (\cos \theta + i \sin \theta)^{-n} \right]$$

$$= \frac{(-1)^n (n-1)!}{2i r^n} \left[\cos n\theta + i \sin n\theta - (\cos n\theta - i \sin n\theta) \right]$$

$$= \frac{(-1)^n (n-1)!}{2i r^n} \cdot 2i \sin n\theta$$

$$= \frac{(-1)^n (n-1)!}{r^n} \sin n\theta \quad \left[\frac{(-1)^n (n-1)!}{(x^2+1)^{n/2}} \sin n\theta \right]$$

$$r = \sqrt{x^2+1}$$

$$y = \tan^{-1}\left(\frac{1+x}{1-x}\right) \quad \text{p.t. } y_n = (-1)^n (n-1)! \sin^n \theta \sin n\theta$$

$$\theta = \tan^{-1}\left(\frac{1}{x}\right)$$

$$\tan^{-1}(a) + \tan^{-1}(b) = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$$

$$\tan^{-1}(1) + \tan^{-1}(x) = \tan^{-1}\left(\frac{1+x}{1-x}\right)$$

$$y = \tan^{-1}\left(\frac{1+x}{1-x}\right) = \pi/4 + \tan^{-1}x$$

$$y_1 = 0 + \frac{1}{1+x^2} \quad \text{same as above}$$

$$y = \sin^{-1}\left(\frac{2x}{1+x^2}\right) \quad \left[\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) \right] = \cos^{-1}\left(\frac{x^2-1}{x^2+1}\right)$$

$$y_n = 2(-1)^n (n-1)! \sin^n \theta \sin n\theta \quad \text{where } \theta = \tan^{-1}\left(\frac{1}{x}\right) = \cos^{-1}(\cos 2\theta)$$

$$\rightarrow x = \tan \theta \quad \sin^{-1}\left(\frac{2 \tan \theta}{1+\tan^2 \theta}\right) = \sin^{-1}(\sin 2\theta) = 2\theta = 2 \tan^{-1} x$$

$$y_1 = \frac{2}{x^2+1}$$

$$\frac{1-\tan^2 \theta}{1+\tan^2 \theta} = \cos 2\theta$$

Leibniz's Thm
 $y = uv$

$$(a+b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + \dots + {}^nC_n a^0 b^n$$

$$y_n = (uv)_n$$

$$= {}^nC_0 u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \dots + {}^nC_n u v_n$$

$$= u_n v + n u_{n-1} v_1 + \frac{n(n-1)}{2!} u_{n-2} v_2 + \dots + u v_n$$

$$y = x^2 e^{ax} = \underbrace{e^{ax}}_u \underbrace{x^2}_v$$

$$u_n = a^n e^{ax} \quad v = x^2$$

$$v_1 = 2x$$

$$v_2 = 2$$

$$v_3 = 0$$

$$y_n = a^n e^{ax} (x^2) + n a^{n-1} e^{ax} (2x) + \frac{n(n-1)}{2!} a^{n-2} e^{ax} (2) + 0$$

HW

$$y = x^3 \sin 2x$$

$\downarrow$
 v

|

$$y = x^n \log x \quad \text{p.t.} \quad y_{n+1} = \frac{n!}{x}$$

$$y_1 = x^n \frac{1}{x} + \log x \cdot n x^{n-1}$$

$$= x^{n-1} + \log x \cdot n x^{n-1}$$

$$= \frac{x^n}{x} + n \log x \frac{x^n}{x}$$

$$x y_1 = x^n + n y$$

$$(x y_1)_n = (x^n)_n + (n y)_n \quad \text{--- (1)}$$

$$(x y_1)_n = \left(\underset{u}{y_1} \underset{v}{x} \right)_n = u_n v + n u_{n-1} v_1 \\ = y_{n+1} x + n y_n (1) \quad \text{--- (2)}$$

$$u = x^n$$

$$v_1 = n x^{n-1}$$

$$v_2 = n(n-1) x^{n-2}$$

$$v_n = n(n-1) \dots (n-(n-1)) x^{n-n} = (n \cancel{n-1} \dots 1) x^{n-n} = n! \quad \text{--- (3)}$$

from (1)

$$y_{n+1} x + \cancel{n y_n} = n! + \cancel{n y_n}$$

$$\Rightarrow y_{n+1} = \frac{n!}{x}$$

$$\frac{x \log(x+1)}{x+1} = \frac{x \log(x+1)}{x+1}$$

$$\begin{aligned} u &= y_1 & v &= x \\ u &= y_2 & v_1 &= 1 \\ u_2 &= y_3 & v_2 &= 0 \\ &\vdots & & \\ u_n &= y_{n+1} & v_n &= 0 \end{aligned}$$

$$y = \cos^{-1} x \quad \text{p.t.} \quad (1-x^2)y_{n+2} - (2n+1)y_{n+1} - n^2 y_n = 0$$

$\downarrow$
 y_2

$\downarrow$
 y_1

$\downarrow$
 y

$$y = \frac{-1}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2} \quad y = -1$$

Diff. w.r.t. x

$$\sqrt{1-x^2} \quad y_2 + y_1 \frac{1}{x\sqrt{1-x^2}} (\cancel{-2x}) = 0$$

$$(1-x^2)y_2 - x y_1 = 0$$

By L.T.H

$$(uv)_n = u_n v + n u_{n+1} v_1 + \frac{n(n+1)}{2!} u_{n+2} v_2 + \dots \rightarrow uv_n$$

$$[(1-x^2)y_2]_n - [x y_1]_n = 0$$

$$[y_2 (1-x^2)]_n - [y_1 x]_n = 0$$

$$y_{n+2}(1-x^2) + n y_{n+1}(-2x) + \frac{n(n+1)}{2!} y_n(\cancel{-2})$$

$$- [y_{n+1} x + n y_n (1)] = 0$$

$$y_{n+2}(1-x^2) - \underbrace{2x n y_{n+1}} - n^2 y_n + \cancel{n y_n} - \underbrace{x y_{n+1}} - \cancel{n y_n} = 0$$

$$(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - n^2 y_n = 0 \quad \checkmark$$

$$y = (x + \sqrt{a^2 + x^2})^m$$

P.T. $a^2 y_{n+2} + (n^2 - m^2) y_n = 0$ at $x=0$.

$$y = m(x + \sqrt{a^2 + x^2})^{m-1} \left(1 + \frac{x}{\sqrt{a^2 + x^2}}\right) y_2 \quad y_2 = y.$$

$$y = m(x + \sqrt{a^2 + x^2})^{m-1} \left[\frac{\sqrt{a^2 + x^2} + x}{\sqrt{a^2 + x^2}} \right]$$

$$\sqrt{a^2 + x^2} y = m(x + \sqrt{a^2 + x^2})^m$$

$$\sqrt{a^2 + x^2} y = m y \quad \text{--- (1)}$$

Diff. w.r.t x

$$\sqrt{a^2 + x^2} y_2 + y_1 \frac{x}{\sqrt{a^2 + x^2}} = m y_1$$

$$(a^2 + x^2) y_2 + x y_1 = m y_1 \sqrt{a^2 + x^2}$$

$$(a^2 + x^2) y_2 + x y_1 = m(m y) \quad \text{from (1)}$$

P.T. L.T.

$$[y_2 (a^2 + x^2)]_n + [y_1 x]_n = m^2 y_n$$

$$y_{n+2} (a^2 + x^2) + n y_{n+1} (2x) + \frac{n(n-1)}{2!} y_n (x^2)$$

$$+ y_{n+1} x + n y_n (1) = m^2 y_n$$

$$y_{n+2} (a^2 + x^2) + 2 x n y_{n+1} + n^2 y_n - n y_n + x y_{n+1} + y_n$$

$$y_{n+2} (a^2 + x^2) + (2n+1) x y_{n+1} + (n^2 - m^2) y_n = 0 = m^2 y_n$$

put $x=0$

$$a^2 y_{n+2} + 0 + (n^2 - m^2) y_n = 0$$

$$y = e^{m \cos^{-1} x} \quad \text{P.T. } (1-x^2) y_{n+2} - (2n+1) y_{n+1} + (n^2 + m^2) y_n = 0$$

$$y_1 = e^{m \cos^{-1} x} \left(-m \frac{1}{\sqrt{1-x^2}} \right)$$

$$\sqrt{1-x^2} y = y (-m) \quad \text{--- (1)}$$

Diff. w.r.t x

$$\sqrt{1-x^2} y_2 + y_1 \frac{-x}{\sqrt{1-x^2}} = -m y_1$$

$$(1-x^2) y_2 - x y_1 = -m y_1 \sqrt{1-x^2}$$

$$(1-x^2) y_2 - x y_1 = -m (-m y) \quad \text{--- (1)}$$

$$y = \sec^{-1} x \quad \text{P.T. } x(x^2-1) y_{n+2} + [(2+3n)x^2 - (n+1)] y_{n+1} + n(3n+1) x y_n + n^2 (n+1) y_{n-1} = 0$$

$$y = \sin(m \sin^{-1} x) \quad / \quad m \sin^{-1} x = \sin^{-1} y$$

$$P.T. \quad (1-x^2) y_{n+2} - (2n+1)x y_{n+1} + (m^2 - n^2) y_n = 0$$

Hence deduce $y_n(0) = 0$ if n is even
 $= (n^2 - m^2) \dots (3^2 - m^2)(1^2 - m^2)m$
 if n is odd.

$$y = \cos(m \sin^{-1} x) \left(m \frac{1}{\sqrt{1-x^2}} \right)$$

$$\sqrt{1-x^2} y_1 = m \cos(m \sin^{-1} x) \quad \text{--- (1)}$$

Differentiate w.r.t x

$$\sqrt{1-x^2} y_2 + y_1 \frac{1}{2\sqrt{1-x^2}} (-2x) = -m \sin(m \sin^{-1} x) \frac{m}{\sqrt{1-x^2}}$$

$$(1-x^2) y_2 - x y_1 = -m^2 y$$

By taking n th order derivative

$$[y_2(1-x^2)]_n - (y_1 x)_n = -m^2 y_n$$

$$(uv)_n = u_n v + n u_{n-1} v_1 + \frac{n(n-1)}{2!} u_{n-2} v_2 + \dots + u v_n$$

$$\Rightarrow y_{n+2}(1-x^2) + n y_{n+1}(-2x) + \frac{n(n-1)}{2!} y_n(-2) - [y_{n+1}(x) + n y_n]$$

$$y_{n+2}(1-x^2) - 2x n y_{n+1} - n^2 y_n - 2 y_n - x y_{n+1} - n y_n = -m^2 y_n$$

$$\checkmark (1-x^2) y_{n+2} - x y_{n+1}(2n+1) + (m^2 - n^2) y_n = 0 \quad \text{--- (2)}$$

$$y(0) = 0 \quad \text{at } x=0$$

$$y_1(0) = m$$

put $x=0$ in (2)

$$y_{n+2}(0) - 0 + (m^2 - n^2) y_n(0) = 0$$

$$y_{n+2}(0) = -(m^2 - n^2) y_n(0) \Rightarrow y_{n+2}(0) = (n^2 - m^2) y_n(0)$$

for $n=0$ $y_2(0) = (2^2 - m^2) y_0(0) = 0 \quad \checkmark$

$n=1$ $y_3(0) = (1^2 - m^2) y_1(0)$
 $= +m(1^2 - m^2)$

$n=2$ $y_4(0) = (2^2 - m^2) y_2(0) = 0 \quad \checkmark$

$n=3$ $y_5(0) = (3^2 - m^2) y_3(0)$
 $= (3^2 - m^2)(1^2 - m^2)m$

$\therefore y_n(0) = 0$ for n is even
 $\& y_n(0) = [(n^2 - m^2)] \dots [(3^2 - m^2)(1^2 - m^2)m$

$$\text{of } \sin^{-1}\left(\frac{y}{b}\right) = \log\left(\frac{x}{n}\right)^n$$

P.T. $x^2 y_{n+2} + (2n+1)x y_{n+1} + 2n^2 y_n = 0$

$$\frac{y}{b} = \sin\left(n \log\left(\frac{x}{n}\right)\right)$$

$$y = b \sin\left(n \log\left(\frac{x}{n}\right)\right)$$

$$y_1 = b \cos\left(n \log\left(\frac{x}{n}\right)\right) \cdot \frac{1}{x/n} \left(\frac{1}{n}\right)$$

$$= \frac{b n}{x} \cos\left(n \log\left(\frac{x}{n}\right)\right)$$

$$x y_1 = b n \cos\left(n \log\left(\frac{x}{n}\right)\right) \quad \text{--- (1)}$$

Diff. wrt. x

$$x y_2 + y_1 = -n b \sin\left(n \log\left(\frac{x}{n}\right)\right) \cdot \frac{1}{x/n} \cdot \frac{1}{n}$$

$$x y_2 + y_1 = -\frac{n^2}{x} y \Rightarrow (x^2 y_2)_n + (x y_1)_n = -n^2 y_n$$

(HW)

(HW)

$$y = \sin\left[\log(x^2 + 2x + 1)\right] = \sin[2 \log(x+1)]$$

P.T. $(x+1)^2 y_{n+2} + (2n+1)(x+1) y_{n+1} + (n^2 + 4) y_n = 0$

$$y = \sec^{-1} x$$

P.T. $x(x^2-1)y_{n+2} + [(2+3n)x^2 - (n+1)]y_{n+1} + n(3n+1)xy_n + n(n^2-1)y_{n-1} = 0$

→ $y_1 = \frac{1}{x\sqrt{x^2-1}}$

$x\sqrt{x^2-1} y_1 = 1 \Rightarrow \sqrt{x^4-x^2} y_1 = 1$
 $\Rightarrow$ ^{Diff wrt x} $\frac{1}{\sqrt{x^4-x^2}} y_2 + y_1 \frac{1}{2\sqrt{x^4-x^2}} (4x^3-2x) = 0$

$(x^4-x^2) y_2 + y_1(2x^3-x) = 0$
 $\cancel{x(x^3-x)} y_2 + \cancel{x} y_1(2x^2-1) = 0$

$\left[\underset{u}{y_2} \underset{v}{(x^3-x)} \right]_n + \left(\underset{u}{y_1} \underset{v}{(2x^2-1)} \right)_n = 0$

① If $x = \sin \theta$ $y = \sin 2\theta$
 p.T. $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - (n^2-4)y_n = 0$



$$y = \sin 2\theta = 2 \sin \theta \cos \theta$$

$$x = 2x \sqrt{1-x^2}$$

$$y = 2x \sqrt{1-x^2}$$

② $x = e^t$ & $y = \cos mt$ p.T.
 $x^2 y_{n+2} + (2n+1)x y_{n+1} + (m^2+n^2)y_n = 0$



$$y = \cos mt$$

$$= \cos m(\log x)$$

$$\text{as } x = e^t$$

$$t = \log x$$

③ $x = \cos \theta$ $\theta = \frac{1}{m} \log y$
 p.T. $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - (n^2+m^2)y_n = 0$

$$\log y = m\theta \Rightarrow y = e^{m\theta}$$

$$y = e^{m \cos^{-1} x}$$

